# Beyond PII Removal: Risk-Budgeted Generalization for LLM-Ready Sensitive Text

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## Abstract

Sensitive text corpora are valuable for training, retrieval, evaluation, and post-training adaptation of foundation-model systems, but common transformation pipelines still focus on direct personally identifying information (PII) such as names, emails, phone numbers, and addresses. This span-level view is necessary but incomplete: a record with no explicit name may still reveal sensitive attributes through quasi-identifiers such as exact dates, fine-grained locations, rare occupations, schools, medical or legal facts, family relations, citizenship cues, or combinations of ordinary details. We introduce Risk-Budgeted Quasi-Identifier Generalization (RB-QIG), a lightweight and auditable transformation layer that scores quasi-identifiers by privacy risk and utility importance, applies the least destructive step in a pre-specified generalization ladder, and stops when a document-level risk budget is met. On a controlled synthetic benchmark, RB-QIG balanced reduces risk-weighted leakage to 23.6% while preserving 71.7% of utility facts; blanket quasi-identifier redaction removes leakage but preserves only 43.3% of utility facts. On a 100-record English RAT-Bench pilot, direct redaction leaves 78.0% LLM-attacker risk-weighted leakage and a naive LLM sanitizer leaves 46.0%, while RB-QIG balanced reduces leakage to 5.7%. We conclude that responsible data transformation for foundation-model pipelines should measure what capable models can still infer, not only which explicit PII spans were removed.

## 1 Introduction

Sensitive text is central to many data sources that would make foundation-model systems more useful: clinical notes, legal decisions, financial records, support logs, internal workplace messages, student advising records, and user-assistant conversations. Such data can support retrieval-augmented generation, model evaluation, safety auditing, preference-data construction, and domain adaptation. The same data also contains details that people did not intend to make public. Responsible data enablement therefore often begins with de-identification: remove names, replace emails, mask phone numbers, and redact addresses.

This direct-identifier pipeline is important, but it is not sufficient for LLM-ready text. A sentence such as "a 34-year-old oncology infusion pharmacist at a small startup in Asheville called about a brother's rare condition after an April 2025 surgery" may contain no name or email after redaction, yet still expose a highly specific profile. Traditional privacy work formalized this problem through linkage and quasi-identifiers Sweeney (2002); Machanavajjhala et al. (2007); Li et al. (2007); Narayanan & Shmatikov (2008). Modern language models make the problem more acute because an attacker need not match exact database columns: LLMs can combine paraphrase recognition, world knowledge, and contextual inference to recover sensitive attributes Staab et al. (2024). Recent text-anonymization benchmarks similarly argue that anonymization should be evaluated by residual re-identification risk, not only by span recall Krčo et al. (2026).

The practical question is therefore not just "did the tool remove obvious PII?" but "what can a capable model still infer, and how much useful meaning was preserved?" A conservative answer is to redact every quasi-identifier: occupations become `[OCCUPATION]`, locations become [LOCATION], dates become [DATE], and diagnoses become [MEDICAL]. This can reduce leakage, but it also destroys much of the information that made the corpus useful. The opposite extreme, direct PII removal only, preserves utility but leaves large residual inference risk.

We study the space between these extremes through Risk-Budgeted Quasi-Identifier Generalization (RB-QIG). RB-QIG does not train a model and does not rewrite whole documents. It takes candidate quasi-identifier spans, assigns each a privacy-risk score and a utility-importance score, and applies span-level replacements from a small ladder: keep, generalize once, generalize further, or redact. The algorithm greedily chooses the next edit that gives the largest risk reduction per unit of utility loss, stopping when a document-level risk budget is satisfied. The resulting audit log records which fact was generalized, which replacement was used, and why.

Our contributions are:

- a quasi-identifier taxonomy for LLM-ready sensitive text;
- a risk-budgeted generalization policy that operationalizes an inspectable privacy–utility tradeoff without training or fine-tuning;
- an evaluation protocol combining deterministic leakage metrics, LLM attackers, and LLM utility judges; and
- an analysis showing that residual leakage often comes from relational or lexical implications rather than missed exact spans.

We show that direct PII redaction and naive LLM sanitization leave substantial residual quasi-identifier risk, while RB-QIG sharply reduces that measured risk and exposes a useful controlled privacy–utility frontier.

## 2 Background and Threat Model

**From tabular quasi-identifiers to text.** Classical privacy work showed that attributes such as ZIP code, gender, and birth date can link a record to a person even when names are absent Sweeney (2002). $l$-diversity and $t$-closeness refined this view by showing that indistinguishability alone may not protect sensitive attributes Machanavajjhala et al. (2007); Li et al. (2007). Differential privacy offers a different guarantee for statistical releases Dwork et al. (2006), but it does not directly solve the instance-level text-transformation problem studied here: preserve a useful text record while reducing inferable identifiers inside that record.

**Text de-identification.** Applied de-identification systems often target named entities and regulated identifier categories. Clinical de-identification has a long history of shared tasks and neural sequence-labeling systems Uzuner et al. (2007); Dernoncourt et al. (2017). Practical tools such as Presidio support PII detection, masking, and replacement Microsoft (2026); specialized privacy filters provide another local PII-masking building block OpenAI (2026). These systems are useful, but they are typically optimized for identifiable spans rather than the residual inference risk created by combinations of demographic, relational, narrative, and domain facts. TAB provides a legal-domain anonymization benchmark Pilán et al. (2022), and RAT-Bench directly emphasizes LLM-based re-identification risk from direct and indirect identifiers Krčo et al. (2026).

**LLM privacy leakage.** LLMs introduce two complementary privacy concerns. Training data may be memorized or extractable under some conditions Carlini et al. (2021; 2023). Separately, even when a private fact is not explicit, a model may infer it from surrounding text. Staab et al. show that LLMs can infer personal attributes from ordinary text in their setting Staab et al. (2024). RB-QIG focuses on the second concern at the transformation stage: before using sensitive text for retrieval, evaluation, or post-training, reduce what a capable model can infer from the transformed record.

Table 1: Quasi-identifier categories used by RB-QIG. The examples are illustrative; the implementation stores type, risk, utility, and a replacement ladder for each candidate span.

| Category | Example precise value | Safer generalized value |
|---|---|---|
| Date/age | April 17, 2025; 34 | month/year; 30s |
| Location | Asheville, NC | city in the southeastern U.S. |
| Occupation | oncology infusion pharmacist | clinical pharmacy professional |
| Institution | named school or employer | university; healthcare company |
| Medical/legal fact | rare condition or case event | broader condition/event type |
| Family relation | younger brother under guardianship | dependent family member |
| Demographic cue | citizenship or heritage phrase | broader demographic placeholder |
| Narrative cue | unique life event | coarse role-preserving description |

**Threat model.** The attacker receives transformed text and, in the LLM-attacker setting, a list of benchmark attributes to infer. The attacker returns structured guesses with short evidence. We report exact recovery, coarse recovery, record compromise, and risk-weighted leakage. We use only synthetic profiles and public benchmarks; we do not attempt to identify real people. This threat model does not cover arbitrary auxiliary databases or determined human investigators. It is a reproducible stress test for whether a transformation leaves sensitive attributes inferable.

We distinguish three evaluation settings. In the *oracle/controlled* setting, ground-truth quasi-identifier spans are known from synthetic data. In the *target-aware public* setting, RAT-Bench target attributes are used to find value-revealing spans; this isolates transformation quality when the sensitive attributes of interest are known. In the *blind public* setting, the extractor does not see target attributes at extraction time and is scored against benchmark annotations afterward. The blind setting is closest to deployment.

# 3 Risk-Budgeted Quasi-Identifier Generalization

## 3.1 Taxonomy and Generalization Ladders

RB-QIG uses typed candidate spans. The taxonomy is intentionally broad because LLM inference often uses facts that are not direct PII. Table 1 summarizes the categories used in the prototype.

A generalization ladder is a short ordered list of replacements. For example, "Boise, Idaho" may map to "a city in the U.S. Mountain West," then "a city in the western United States," and finally `[LOCATION]`. The ladder is not learned; it is a transparent policy artifact. This matters for sensitive data owners because they may want to inspect, revise, or prohibit particular replacement types before releasing transformed text.

## 3.2 Risk and Utility Scores

Each quasi-identifier $q_i$ receives a privacy-risk score $r_i \in [0, 5]$ and a utility-importance score $u_i \in [0, 5]$. Risk reflects specificity, rarity, sensitivity, and linkability. Utility reflects whether the fact is needed for the benign task: diagnosis category may be important in a clinical triage note, while an exact date of birth is usually not. The current prototype uses transparent heuristics and benchmark metadata rather than learned risk models. This is a limitation, but it also makes the method auditable.

The document risk score is

$$R(x) = \sum_i r_i + 0.5 \binom{|\{i : r_i \geq 3\}|}{2},$$

where the second term approximates combination risk from multiple linkable high-risk attributes. Candidate edits are evaluated by their risk drop divided by utility loss. RB-QIG greedily applies the edit with the best ratio until $R(x)$ falls below the budget or no edits remain. All edits are span replacements, so the system avoids unconstrained paraphrase hallucination.

### 3.3 Baselines

We compare against five baselines. *None* leaves the record unchanged. *Direct redaction* removes direct identifiers using regexes and benchmark-provided direct spans. *Naive LLM direct* asks an LLM sanitizer to remove personally identifying information while preserving meaning. *Blanket QI redaction* replaces every quasi-identifier with a typed placeholder. *Presidio-style pattern redaction* is a practical pattern baseline included as a lower-bound diagnostic rather than a full production Presidio deployment. RB-QIG is evaluated with strict, balanced, and utility-oriented budgets in the synthetic setting, with the balanced policy as the main public method.

## 4 Experiments

**Synthetic control.** We generate 100 synthetic records across medical, legal, financial, workplace, and education-support domains. Each record has direct identifiers, quasi-identifiers, ground-truth attributes, a task label, and utility facts. Because oracle quasi-identifier spans are known, this setting isolates the transformation tradeoff: given the sensitive spans, does risk-budgeted generalization preserve more useful information than blanket redaction?

**RAT-Bench pilot.** We evaluate 100 English difficulty-1 RAT-Bench records Krčo et al. (2026). The target-aware version uses benchmark target attributes to identify value-revealing spans and variants. We also run a blind extractor over the same 100 records and score it against benchmark quasi-identifiers. A deterministic backstop for common demographic, citizenship, education, employment, marital, and race/ethnicity cues improves coverage.

**TAB legal screen.** We run a 30-document TAB ECHR legal-domain screen Pilán et al. (2022). The TAB experiment is intentionally bounded and deterministic. It tests whether the residual-risk story transfers to another domain.

**LLM attacker and utility judge.** For RAT-Bench, an LLM attacker receives transformed text and requested attribute names, then returns structured guesses with evidence. An LLM utility judge scores label preservation, fact preservation, and semantic utility without rewarding preserved direct identifiers. We also run bounded stronger-attacker checks with GPT-5.5 and a mini-model agreement check on the blind public outputs. All API calls use synthetic or public benchmark records.

**Metrics.** Privacy metrics are direct-identifier leakage, exact attribute leakage, coarse attribute leakage, record compromise, and risk-weighted leakage. Utility metrics are utility-fact preservation, label preservation, semantic utility, annotation-derived quasi-identifier specificity, and token change.

## 5 Results and Analysis

**Controlled privacy–utility frontier.** Table 2 shows the cleanest evidence for the method. Direct redaction preserves utility but leaves all quasi-identifiers. Blanket quasi-identifier redaction eliminates measured leakage but preserves only 43.3% of utility facts. RB-QIG balanced preserves 71.7% of utility facts while reducing risk-weighted leakage from 100.0% to 23.6%. This supports the controlled claim that risk-budgeted generalization can occupy a useful middle point between direct redaction and blanket redaction.

**Semantic utility is more modest.** The LLM utility judge in Table 4 is more forgiving than the synthetic fact metric because broad placeholders can still preserve some document

Table 2: Synthetic controlled benchmark.

| Method | Risk-weighted leak | Utility facts | Token change |
|---|---|---|---|
| None | 100.0 | 100.0 | 0.0 |
| Direct | 100.0 | 100.0 | 13.5 |
| Blanket QI | 0.0 | 43.3 | 63.7 |
| RB-QIG strict | 16.1 | 48.3 | 59.3 |
| RB-QIG balanced | 23.6 | 71.7 | 59.0 |
| RB-QIG utility | 38.0 | 95.0 | 56.6 |

Table 3: RAT-Bench LLM attacker results.

| Method | Record compromise | Exact attr leak | Coarse attr leak | Risk-weighted leak |
|---|---|---|---|---|
| Direct | 76.0 | 74.9 | 81.2 | 78.0 |
| LLM direct | 34.0 | 48.7 | 52.8 | 46.0 |
| Blanket QI | 3.0 | 3.9 | 7.3 | 5.3 |
| RB-QIG balanced | 2.0 | 4.0 | 8.8 | 5.7 |

meaning. Under this metric, RB-QIG balanced has only a modest semantic utility edge over blanket redaction: +2.6 points. Local placeholder, fluent-rewrite, and attribute-level fluent diagnostics remove surface artifacts or improve specificity, but do not rescue this public-utility gap. A text-quality audit further suggests that phrase-level surface cleanup is not the missing ingredient: the lowest-artifact attr-fluent variant has the worst hidden-label utility, and longer or more generalized replacements correlate negatively with utility in the pooled audit. A same-text audit also finds that the method-labeled privacy-aware judge is noisy for fine-grained rankings. A corrected method-blind, deduplicated rerun removes these same-text inconsistencies but remains negative: RB-QIG balanced minus blanket is -4.0 points, and fluent minus blanket is -5.2 points. We therefore scope the large utility claim to controlled synthetic facts and report public utility as an unresolved limitation.

**Direct redaction leaves high residual LLM risk.** Table 3 shows that direct redaction is insufficient on the RAT-Bench pilot. After direct redaction, the LLM attacker still recovers many target attributes: 78.0% risk-weighted leakage. The naive LLM sanitizer improves over direct redaction but still leaves 46.0% leakage. RB-QIG balanced reduces leakage to 5.7%, a paired drop of 72.2 points relative to direct redaction and 40.2 points relative to the LLM sanitizer. Thus the privacy claim is reduction relative to direct and naive redaction, not superiority over blanket redaction.

**Blind extraction is the deployment bottleneck.** Table 5 summarizes the blind public stress test. The raw blind extractor covers 72.8% of benchmark QI spans; a generic deterministic backstop raises coverage to 99.6% . Under the LLM attacker, blind-backstop RB-QIG balanced reduces risk-weighted leakage from 78.1% for direct redaction to 6.4%, and is statistically tied with blanket redaction at -0.5 points. Deterministic blind scoring gives the same direct-redaction failure but is harsher on generalized cues: direct redaction leaves 94.3%, RB-QIG balanced leaves 5.4%, the direct-to-RB-QIG drop is 88.9 points, and RB-QIG remains 2.3 points leakier than blanket. Public semantic utility is also tied with blanket redaction (-0.4 semantic-utility points), so the blind setting should be read as evidence for residual-risk reduction and as evidence that robust utility-preserving quasi-identifier extraction and rewriting remain open.

**Stronger attackers and budget diagnostics.** A 50-record GPT-5.5 attacker check is harsher than the low-cost attacker: direct redaction leaves 87.8% risk-weighted leakage, RB-QIG balanced leaves 31.9%, and blanket QI leaves 29.7%. The direct-to-RB-QIG reduction remains large. A 20-record blind mini-model check gives the same qualitative ordering: direct 75.0%, RB-QIG balanced 14.8%, and blanket QI 16.8%. Pairwise exact/coarse leak agreement is moderate: 83.3%/80.0% for mini versus GPT-5.5, and 76.7%/71.9% for nano versus GPT-5.5, so attacker strength affects absolute leakage but not the qualitative ordering. budget-frontier

Table 4: Synthetic LLM utility judge results.

| Method | Label preserved | LLM fact preservation | Semantic utility |
|---|---|---|---|
| Direct | 100.0 | 75.3 | 83.2 |
| Blanket QI | 99.0 | 69.1 | 76.8 |
| RB-QIG balanced | 100.0 | 71.3 | 79.4 |

Table 5: Blind public RAT-Bench stress test.

| Method | LLM risk-weighted leak | Semantic utility | LLM fact preservation |
|---|---|---|---|
| Direct | 78.1 | 86.2 | 84.3 |
| Blanket QI | 6.8 | 62.6 | 53.8 |
| RB-QIG balanced | 6.4 | 62.2 | 55.5 |

diagnostics show the control knob is meaningful: on blind RAT-Bench, budgets 2/4/6 give 4.4%/5.4%/7.2% deterministic risk-weighted leakage and 30.1%/33.2%/36.4% quasi-identifier specificity; on TAB they give 7.8%/12.3%/20.4% leakage and 28.1%/31.2%/34.3% specificity. Annotation-derived specificity is higher for RB-QIG balanced than blanket by +6.8 points on blind RAT-Bench and +6.2 on TAB. Additional target-aware smokes preserve the same ordering on difficulty-2 (direct 70.8% and naive sanitizer 56.9% versus RB-QIG 13.1%) and on the 20-record target-aware mini smoke (RB-QIG 18.0% versus 77.3% for direct).

**Cross-domain and practical-baseline diagnostics.** The TAB screen provides deterministic cross-domain support: direct redaction leaves 99.8% risk-weighted leakage, while RB-QIG balanced lowers it to 12.3%. Its bounded 20-document legal-task utility screen shows that blanket scores 73.0%, while RB-QIG balanced scores 71.0%. The Presidio-style pattern-only baseline gives a practical lower-bound diagnostic: on RAT-Bench it leaves 37.0% direct-identifier leakage and 85.0% deterministic risk-weighted leakage; on TAB it misses legal names and case identifiers, leaving 100.0% direct-identifier leakage. This makes the oracle-assisted direct baseline conservative rather than weak.

**Failure taxonomy.** Residual RB-QIG leaks are mostly relational or lexical variants: widowhood from bereavement cues, sex from gendered language, education from school-name mentions, citizenship from non-citizen phrasing, and employment or armed-forces status from occupation context. These failures are not just missed exact spans; they are cues that a model can combine. This supports the broader argument that data transformation should evaluate inferability, not only span categories.

## 6 Limitations and Ethics

RB-QIG is not an anonymization guarantee and should not be used as a legal or compliance claim. The main public benchmark extraction is target-aware, so it evaluates transformation behavior under known benchmark attributes rather than fully blind deployment. All experiments use synthetic or public benchmark data. We do not attempt to identify real people or infer private attributes about individuals.

## 7 Conclusion

Direct PII removal is an incomplete transformation for LLM-ready sensitive text. RB-QIG demonstrates that a small, auditable risk-budgeted generalization layer can dramatically reduce measured residual inference risk while preserving useful semantic content better than blanket redaction in controlled settings. The main open problem is robust blind quasi-identifier extraction and rewriting that preserves task-relevant utility while handling relational, lexical, and world-knowledge inference cues. More broadly, responsible data

Table 6: Representative residual failure categories.

| Failure type | Example cue | Why span removal misses it | Possible fix |
|---|---|---|---|
| Relation cue | family status or bereavement phrase | sensitive status is inferred from a relation, not named directly | relation-aware rewrite |
| Gender cue | pronouns or gendered family roles | sex can be inferred after direct identifiers are removed | neutralization pass |
| Cultural cue | heritage, tradition, or language mention | race or ethnicity may be inferred from ordinary context | coarser cultural phrase |
| Institution cue | school, employer, or service branch | education or employment is exposed by a named institution | institution ladder |
| Placeholder cue | typed demographic placeholder in context | the placeholder itself can reveal the attribute class | neutral placeholders |

enablement should evaluate what capable models can still infer, not only what explicit PII spans were removed.

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